RAG Triad Evaluations – Claude LLM Judge

A. Scenario 1

1. Architectural Faithfulness Audit

- **Score:** 4/5
- **Relevant Passage(s):** "the total absence of data in the Last Name field (0% population)" and, in Evidentiary Adequacy, "verified by the Last Name field, which contains 0 non-null values across the entire 50-row sample."
- **Evaluation basis:** KBS 4 states that Last Name "is almost universally empty across all entries" — a hedged, near-total pattern, not an absolute one. Faithfulness rubric anchor 4 ("all major pattern claims match... at most one minor claim is imprecisely scoped").
- **Justification:** The overwhelming majority of the text's structural-silence claims (Last Name/First Name asymmetry, Sale Date/Sale Disposition prospective silence, Skills Specified concentration) track KBS 4 closely, including appropriate causal framing (commodification, legal denial of patronymics). However, the text's shift from KBS's qualified "almost universally empty" to an unqualified "0% population" / "0 non-null values" overstates certainty on a central claim, which is the kind of minor scope inflation the rubric penalizes at the 4-point level rather than the 5-point level. I flag as an **evaluation limitation** that the specific numeric claims (37 non-null FULL TEXT, 36 Notes, 7 Count, 17/50 Sale Date, 45 Subscriber State, 21 Sale Location County, 25 Age Specified, 24 Skills Specified, and the row 5/row 7 date discrepancies) are dataset-level statistics not addressed qualitatively by KBS 1–4 and therefore cannot be confirmed or denied against the supplied ground truth; they are not scored as either faithful or unfaithful.

2. Architectural Context Recall - Primary Corpus Integration

- **Score:** 4/5
- **Relevant Passage(s):** Section (6) "Structured Absences, Silences, and Uncertainties" (tripartite typology: Accidental/Technical, Practical, Structural); Section (5) spatial/temporal treatment of County, Sale Location fields, and Ad Date/Press Date divergence; Section (3) Translational Adequacy discussion of Entity Resolution unreliability.
- **Evaluation basis:** KBS 1 (representational content), KBS 2 (spatial/temporal), KBS 3 (entity resolution baseline), KBS 4 (structural silences).

- **Justification:** KBS 1, 2, and 4 are integrated thoroughly and connected to analytical consequences rather than merely listed (e.g., the silence typology is tied to computational handling guidance). KBS 3 is engaged only at the level of methodological consequence (entity resolution is "unreliable... due to systemic anonymization") but omits the specific corpus-level findings KBS 3 establishes — the mix of named individuals (e.g., patterns of naming in legal vs. private contexts) and the "bulk demand"/family-group anonymity patterns. This is a single significant omission against an otherwise strong integration of the other three KBS sources, consistent with the 4-point anchor. Note per scoring guidance: Scenario 1 has no retrieval mechanism — the primary corpus (Notebook 1 equivalent) was supplied directly in context rather than retrieved, so this score reflects integration fidelity only, not retrieval performance.

3. Architectural Context Recall - Sociocultural Context

- **Score:** 2/5
- **Relevant Passage(s):** "Statutes dictated public notice requirements for probate sales, sheriff sales, executions for debt, and the disposal of freedom-seeking individuals under fugitive slave laws (Sale Disposition). The differentiation of transaction types (Ad Type...) reflects distinct legal and economic practices, where hiring out served as a flexible method to extract surplus value without transferring long-term capital ownership..."
- **Evaluation basis:** KBS 5 (record-generating purpose, legal structures — specific statutes: 1719/1792/1803 runaway laws, 1832 Act, 1796 Importation Ban, 1818 Exportation Act, 1835 hiring act); KBS 6 (actors/omissions); KBS 8 (spatial/temporal economic drivers); KBS 10 (historiography).
- **Justification:** The text explicitly discloses that "no external secondary context notebook (Notebook 2) was provided," and this is borne out in the content: legal and economic claims ("fugitive slave laws," "probate sales," generic reference to hiring-out economics) are generic enough that they cannot be checked against the specific statutory apparatus KBS 5 documents (no dates, act names, or mechanisms cited), nor does the text draw on the economic drivers in KBS 8 (Panic of 1819, cotton-belt price inflation) or the historiographical apparatus in KBS 10 (Fogel, Baptist, Berry, Fields, etc.). The one instance where legal context is tied to a specific field (Ad Type/Sale Disposition) is a genuine but non-specific and non-checkable gesture toward KBS 5-type content, placing this just above the 1-point floor but well short of the specific, sourced grounding required for a 3 or higher.

4. Architectural Answer Relevance - Interpretive Synthesis

- **Score:** 5/5
- **Relevant Passage(s):** "Intersectional Data Feature: The intersection of socio-legal notice mandates and structural commodification practices produces a distinct data feature: the concentration of descriptive precision in Skills Specified and Age Specified alongside absolute anonymity in Last Name... is an artifact produced jointly by commercial marketing incentives and oppressive legal structures." Also the "Distinguishing Patterns from Artifacts" table (Section 4), which repeatedly cross-references Subscriber Locale density, Ad Type, and age-rounding conventions to explain three separate computational artifacts.
- **Evaluation basis:** Answer Relevance — Interpretive Synthesis rubric anchor 5 ("connects two or more analytical dimensions to explain a corpus feature that no single dimension explains alone, and does this more than once").
- **Justification:** The text performs explicit, repeated intersectional reasoning rather than additive listing: the Section 1 "Intersectional Data Feature" subsection explicitly names two interacting causal mechanisms producing a single data pattern, and the diagnostic table in Section 4 performs the same move three additional times (skills/geography, gender/ad-type, age-clustering/estimation artifact). This satisfies the "more than once" threshold for the top anchor.

5. Architectural Answer Relevance - Corpus Relevance Audit

- **Score:** 5/5
- **Relevant Passage(s):** Specific field-level and value-level anchoring throughout, e.g., "Ad Date 18280719 versus Press Date 18280726, or Ad Date 18300529 versus Press Date 18301113," and the field-by-field null-count discussion (Last Name, FULL TEXT, Notes, Count, Sale Date, Subscriber State, Sale Location County, Age Specified, Skills Specified).
- **Evaluation basis:** Answer Relevance — Corpus Relevance rubric anchor 5 ("every substantive claim is anchored to this corpus's specific schema... could not be substituted for an analysis of a different, similar dataset").
- **Justification:** Nearly every analytical claim is tied to a named field in the 27-field schema and, where possible, to specific row-level values or null-count statistics unique to this 50-row sample. This level of schema- and value-specific anchoring means the text could not be repurposed as a generic essay on slave-trade advertisements without substantial rewriting, satisfying the top anchor. (As noted in Section 1, the accuracy of the specific counts themselves is not verifiable against the supplied KBS materials — this score addresses specificity/relevance, not independent verification of the underlying figures.)

6. Architectural Performance Summary

- RAG Architecture Total Score: 20/25

Evaluation limitations flagged: (1) Several dataset-level numeric claims (null counts for FULL TEXT, Notes, Count, Sale Date, Subscriber State, Sale Location County, Age Specified, Skills Specified; specific Ad Date/Press Date pairings attributed to "rows 5 and 7") are not addressed by the qualitative KBS 1–4 ground truth and are therefore unscoreable against this rubric — they are neither confirmed nor contradicted here. (2) As a parametric-baseline (Scenario 1) run with no Notebook 2 access, the sociocultural context recall score reflects the model's internal parametric knowledge only, not retrieval performance; this is consistent with the architecture's design and should be read as a ceiling effect rather than a generation failure specific to this run.

B. Scenario 2

A methodological note before scoring: Scenario 2 corresponds to your Condition 2 (single-notebook RAG). The SDB itself flags this explicitly ("No scholarly context notebook (Notebook 2) was provided for this scenario"). Per your documented scoring convention, I am applying the rubric uniformly and treating the resulting low Sociocultural Context score as an **architectural ceiling to be documented**, not evidence of a generation-quality failure — this distinction matters for how you narrate the result in the findings chapter versus how the number reads in isolation.

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1. Architectural Faithfulness Audit

- **Score:** 4/5
- **Relevant Passage(s):** The "Structured Absences" claim that the First Name field "is most frequently populated in 'Sheriff's Sale' or public estate records" (appropriately hedged with "most frequently" rather than "always"); the Actors/Perspectives claim that enslaved perspectives "are entirely excluded from the corpus."
- **Evaluation basis:** KBS 4 (structural silences — near-total absence of Last Name, first names "primarily appear only in legal contexts"); RAG Triad Faithfulness anchors 4–5.
- **Justification:** Most corpus-wide pattern claims track the KBS 1–4 ground truth closely and use appropriately hedged language ("most frequently," "systematically," "an open interpretive question") rather than overclaiming universality. The one point of imprecision is the "entirely excluded" framing for enslaved perspectives — a strong, unhedged claim that is well-supported by KBS 6 (Notebook 2 territory) but is not itself a pattern KBS 1–4 alone establishes; it reads as imported interpretive framing rather than a corpus-derived frequency claim, which is a minor scope slippage rather than a factual contradiction.

2. Architectural Context Recall — Primary Corpus Integration

- **Score:** 5/5
- **Relevant Passage(s):** The "Intersection Account" explicitly linking the legal mandate for property seizure (Dimensions 1–2) to the anonymization pattern (Dimension 6) to explain differential First Name population rates between Sheriff's Sale and private ads.

- **Evaluation basis:** KBS 1 (representational conventions), KBS 2 (spatial/temporal spread), KBS 3 (entity resolution baseline), KBS 4 (structural silences); Context Recall anchor 5.
- **Justification:** All four KBS 1–4 domains are represented and, critically, connected to the analysis rather than merely enumerated — the intersection account is the clearest evidence of integration rather than listing.

3. Architectural Context Recall — Sociocultural Context

- **Score:** 1/5
- **Relevant Passage(s):** "Citation Specificity Note: No scholarly context notebook (Notebook 2) was provided for this scenario; therefore, all evidence is strictly derived from the primary dataset, and no external scholarly citations are asserted."
- **Evaluation basis:** KBS 5–10 (legal statutes, historiography, actors/omissions framework — none engaged); Context Recall anchor 1.
- **Justification:** No KBS 5–10 content (specific statutes, named historiographical sources, or the "soul value"/commodification scholarly apparatus) appears anywhere in the text. This is consistent with the retrieval architecture as designed for this scenario rather than a synthesis failure — the text is explicit and transparent about the boundary rather than fabricating Notebook 2-style content, which is itself methodologically significant and worth noting separately from the numeric score.

4. Architectural Answer Relevance — Interpretive Synthesis

- **Score:** 5/5
- **Relevant Passage(s):** The Intersection Account (legal mandate × anonymization → differential name-field population); the Analytical Utility section's explanation of a hypothetical age-clustering artifact as a product of advertiser rhetorical strategy interacting with market-demand targeting, rather than a demographic fact.
- **Evaluation basis:** Answer Relevance — Interpretive Synthesis anchor 5 (two or more explicit intersectional accounts).
- **Justification:** Two distinct, explicit instances of cross-dimensional reasoning are present, each explaining a corpus feature that no single dimension explains alone — satisfying the top anchor's "more than once" criterion.

5. Architectural Answer Relevance — Corpus Relevance Audit

- **Score:** 5/5
- **Relevant Passage(s):** Specific field citations (Subscriber Firstname/Lastname/Locale/State, Sale Disposition, Sale Location Locale), county-level specificity (Dorchester, Queen Anne's, Baltimore City), literal date-string examples (Ad Date 18300529 vs. Press Date 18301113), and direct phrase citations ("stout healthy NEGRO MAN," "valuable servants").
- **Evaluation basis:** Answer Relevance — Corpus Relevance anchor 5.
- **Justification:** Claims are consistently anchored to this corpus's specific schema and content rather than generalized historical discussion; the text could not be substituted for analysis of a different comparable dataset without substantial rewriting.

6. Architectural Performance Summary

- RAG Architecture Total Score: 20/25

Evaluation limitations flagged, per your instrument's instructions: The record-level accuracy of specific cited names/dates (e.g., whether "Jacob" and "Rachel" are correctly attached to the rows implied) falls outside this rubric's scope and would need separate manual verification. The Sociocultural Context score of 1/5 should be reported alongside the architectural explanation above rather than as a standalone deficiency score — collapsing it into the total without that context risks misrepresenting Condition 2's designed limitation as a generation failure, which is a distinction worth preserving explicitly in your Chapter 4 findings narrative given your stated commitment to documenting architectural ceilings rather than folding them into scoring adjustments.

C. Scenario 3

Acting as the automated judge specified in your prompt, here is the corpus-level evaluation of the Scenario 3 SDB against the supplied KBS 1–10 ground truth.

A methodological note before the scores: per your standing preference for citation rigor, I want to flag that several sources cited *within the SDB text itself* — Christopher Haley's "Legacy of Slavery in Maryland: 'Chattels'" (*Journal of Slavery and Data Preservation*, 2024), Jennifer Hull Dorsey's *Hirelings* (2011), and Stephen D. Behrendt's "Seasonality in the Trans-Atlantic Slave Trade" — do not appear anywhere in the KBS10 historiographical baseline you supplied, nor am I able to independently confirm the Haley journal article

exists (I'm not certain it does; "Journal of Slavery and Data Preservation" is not a title I can verify). Similarly, the specific legislative citations in the SDB (Maryland Act of 1752, Act of 1831 Ch. 323, Acts of Feb. 12 1833 and March 18 1840, with "Record #" numbers) do not match the specific acts enumerated in KBS5 (which lists 1719/1792/1803 runaway statutes, the 1832 Act — not 1831 — Relating to Free Negroes and Slaves, 1839/1840 vagrancy law, 1681 Orphans' Estate Law, 1796 Importation Ban, 1818 Exportation Act, and the 1835 Cross-Border Hiring Act). I am treating this as an *evidentiary traceability* problem rather than double-penalizing it under Faithfulness or Context Recall, consistent with the rubric's instruction not to decompose into atomic fact-checks

— but it materially limits confidence that this content originated in Notebook 2 retrieval rather than parametric knowledge, and you should verify these citations independently before treating them as confirmed.

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1. Architectural Faithfulness Audit

- **Score:** 4/5
- **Relevant Passage(s):** "The fields First Name and Last Name are frequently blank..."; "The fields Sale Date and Sale Disposition are heavily unpopulated"; the claim of "a high concentration of specific vocational attributes recorded in Skills Specified for young males (aged 15–25)."
- **Evaluation basis:** KBS4 (structural silences in Last Name, First Name, Sale Date); Faithfulness rubric anchors 4–5.
- **Justification:** The hedged claims about name and sale-date omission are consistent with KBS4's "near-total absence" and "systemic null" language, and the text does not overstate these into universal claims. However, the specific age-cohort claim (15–25) tying skills concentration to that range is not addressed by KBS1–4 at all — KBS1 mentions a *purchasing* age range (12–25) for one trader's agents, which is a distinct fact. This claim is unscoreable against the supplied ground truth rather than confirmed or contradicted, and is flagged as a limitation rather than averaged into the score.

2. Architectural Context Recall - Primary Corpus Integration

- **Score:** 4/5
- **Relevant Passage(s):** Section 1(4) Representational Conventions; Section 1(5) Spatial and Temporal Variation; Section 1(6) Structured Absences, Silences, and Uncertainties.
- **Evaluation basis:** KBS1 (representational content), KBS2 (spatial/temporal),

KBS4 (structural silences); Context Recall rubric anchor 4.

- **Justification:** KBS1, KBS2, and KBS4 patterns are integrated analytically rather than merely listed — e.g., the Sale Date/Sale Disposition discussion connects directly to KBS4's public-vs-private recording distinction. However, KBS3 (entity resolution — the specific named network of enslaved individuals, owners, and traders such as Woolfolk, Scotti, and named sheriffs) is essentially absent; the SDB's entity-resolution discussion uses a generic placeholder name ("Tom") rather than engaging KBS3's actual content. This constitutes one significant omission, consistent with the score-4 anchor.

3. Architectural Context Recall - Sociocultural Context

- **Score:** 4/5
- **Relevant Passage(s):** Section 1(2), connecting the 1831/1833/1840 legislative acts to advertisers' requirement to "document specific details in the text, such as a clear statement of title or term of service"; Hypothetical Scenario B, connecting seasonal agricultural cycles to an autumn hiring-ad anomaly.
- **Evaluation basis:** KBS5 (legal/economic structures), KBS8 (spatial/temporal variation, seasonal hiring); Context Recall rubric anchor 4–5.
- **Justification:** The text does use interpretive/legal frameworks as explanatory grounding for corpus-level features more than once (legal compliance → textual content requirements; seasonal wheat economy → autumn hiring spike), which is the behavior the score-5 anchor rewards. I am not scoring this a 5, however, because the specific legal instruments cited do not match those enumerated in KBS5, and none of the KBS10 historiographical sources (Fogel, Baptist, Berry, Johnson, Fields, Grivno, Whitman, etc.) are cited anywhere in the SDB — the scholarly citations used instead (Dorsey, Behrendt) are absent from the supplied ground truth. This raises a traceability concern about whether the explanatory content reflects genuine Notebook 2 retrieval, even though the structural pattern of "framework explains corpus feature" is present.

4. Architectural Answer Relevance - Interpretive Synthesis

- **Score:** 5/5
- **Relevant Passage(s):** "Intersection of Dimensions" paragraph explaining the skills-concentration/surname-absence configuration as the product of Dimension 1 (Commercial Function) and Dimension 4 (Representational Conventions); Hypothetical Scenarios A and B in Section 4, each explicitly tracing a computational anomaly to an interacting set of historical/economic conditions.

- **Evaluation basis:** Answer Relevance (Interpretive Synthesis) rubric anchor 5.
- **Justification:** The text explicitly names and combines multiple analytical dimensions to explain a feature that no single dimension explains alone, and does so more than once (the intersection paragraph, plus two independent scenario analyses). This meets the top anchor's requirement of repeated, explicit relational reasoning rather than accumulated disconnected observations.

5. Architectural Answer Relevance - Corpus Relevance Audit

- **Score:** 5/5
- **Relevant Passage(s):** Consistent, granular reference to actual schema fields throughout (FULL TEXT, Skills Specified, Age Specified, Sale Date, Sale Disposition, Subscriber Firstname/Lastname/County, County vs. Sale Location County) and specific Maryland counties (Talbot, Dorchester, Eastern Shore).
- **Evaluation basis:** Answer Relevance (Corpus Relevance) rubric anchor 5.
- **Justification:** Nearly every substantive claim is tied to a named field or a specific geographic/legal detail from this corpus rather than generic slavery-historiography discussion; the text could not be substituted wholesale for an analysis of a different comparable dataset without rewriting.

6. Architectural Performance Summary

- RAG Architecture Total Score: 22/25

Unscoreable/flagged items (not averaged into the above): the 15–25 age-cohort skills claim (Faithfulness); the specific legislative citations and non-KBS10 scholarly citations (Sociocultural Context). Both should be independently verified before this evaluation is treated as a clean pass

— my assessment of "genuine explanatory grounding" is based on structural/rhetorical pattern-matching to the rubric, not on confirming that the underlying citations are real or accurately dated.